A Root-Complete Cnoidal and Soliton Atlas for KdV Traveling Waves

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Abstract

We classify every bounded classical real traveling profile of the plus-sign Korteweg–de Vries equation directly from its cubic first integral. Three simple real roots give one translated Jacobi-cn^2 family with exact speed and fundamental period; a lower double root gives the sech^2 soliton, and all remaining root topologies give only bounded constants. The classification also closes the first two period moments, the harmonic limit, Galilean covariance, and the physical-time return on a fundamental circle. Twelve rational-root rows are checked by independent regularized quadrature and exact symbolic reconstruction.

1 Frozen equation and cubic reduction

We fix

$$u_t + 6uu_x + u_{xxx} = 0, \qquad u(x,t) = U(\xi), \quad \xi = x - ct, \tag{1}$$

with $U \in C^3(\mathbb{R};\mathbb{R})$ bounded. Integrating the traveling equation twice gives

$$U'' = cU - 3U^2 + A, \qquad (U')^2 = -2U^3 + cU^2 + 2AU + B. \tag{2}$$

If the cubic has real roots $r_1 \le r_2 \le r_3$, its compact allowed branch has the convention

$$(U')^2 = 2(r_3 - U)(U - r_2)(U - r_1), \tag{3}$$

and coefficient comparison forces

$$c = 2(r_1 + r_2 + r_3), \quad A = -(r_1r_2 + r_1r_3 + r_2r_3), \quad B = 2r_1r_2r_3. \tag{4}$$

These signs are part of the frozen object: switching the sign of $6uu_x$ would change every speed convention.

2 Complete bounded-profile theorem

Theorem 1. *Every nonconstant bounded entire profile satisfying (2) is, up to translation of ξ, exactly one of the following.*

1. *If $r_1 < r_2 < r_3$, put*

$$m = \frac{r_3 - r_2}{r_3 - r_1}, \qquad k = \sqrt{\frac{r_3 - r_1}{2}}.$$

Then $0 < m < 1$ and

$$U(\xi) = r_2 + (r_3 - r_2)\,\mathrm{cn}^2(k\xi; m). \tag{5}$$

2. *If $r_1 = r_2 < r_3$, then*

$$U(\xi) = r_1 + (r_3 - r_1)\,\mathrm{sech}^2(k\xi), \qquad k = \sqrt{(r_3 - r_1)/2}. \tag{6}$$

An upper double root, a triple root, or a cubic with only one real root has no nonconstant bounded entire profile. Constant profiles remain solutions, but do not select a traveling speed.

Proof. The negative leading coefficient in (2) leaves a compact positive interval only for three simple roots, namely $[r_2, r_3]$, or for the lower-double-root closure $[r_1, r_3]$. Simple endpoints are reached and reflected in finite time; the lower double root is approached only at infinite time. Every other positive component is unbounded, proving exhaustion. For $q = \mathrm{cn}^2(k\xi; m)$,

$$q_\xi^2 = 4k^2 q(1 - q)(1 - m + mq).$$

Substitution of the displayed m and k is exactly (3); its derivative gives the profile ODE. The identity $\mathrm{cn}(s; 1) = \mathrm{sech}\, s$ gives the homoclinic face. Thus the result is a root classification, not an ansatz selected from a numerical plot.

3 Period, moments, degenerations, and clock

Since cn^2 has period $2K(m)$, the *fundamental* spatial period and mean are

$$L = 2\sqrt{\frac{2}{r_3 - r_1}}K(m), \qquad \langle U \rangle = r_1 + (r_3 - r_1)\frac{E(m)}{K(m)}. \tag{7}$$

The common $4K$ convention for cn would double L incorrectly. With

$$C_2 = \frac{E - (1-m)K}{mK}, \quad C_4 = 1 - \frac{2(K-E)}{mK} + \frac{(2+m)K - 2(1+m)E}{3m^2K},$$

one further exact observable is

$$\langle U^2 \rangle = r_2^2 + 2r_2(r_3 - r_2)C_2 + (r_3 - r_2)^2 C_4. \tag{8}$$

As $r_2 \downarrow r_1$, (5) tends to (6); its excess mass is $2\sqrt{2(r_3 - r_1)}$. As $r_2 \uparrow r_3$ with r_1 fixed, the amplitude vanishes and

$$L \longrightarrow \frac{2\pi}{\sqrt{2(r_3 - r_1)}}.$$

For every $a \in \mathbb{R}$, $u_a(x,t) = u(x - 6at, t) + a$ solves (1); all roots shift by a and c by $6a$, while m and L are unchanged. On a circle of length L, $c \neq 0$ gives primitive physical-time return $T = L/|c|$; $c = 0$ is stationary.

4 Independent finite receipt

The producer evaluates twelve ordered rational-root triples at 90 decimal digits. The checker does not infer (7) from those values: it independently computes

$$L = 2\sqrt{2}\int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{r_2 - r_1 + (r_3 - r_2)\sin^2\theta}}$$

and uses the same regularized weight for $\langle U \rangle$ and $\langle U^2 \rangle$. It then checks separate elliptic nodes. The full gate contains

Table 1: Representative rows; JSON retains 75 significant digits.

row	r_1	r_2	r_3	m	c	L
P01	-5	-2	1	1/2	-12	2.1409
P06	-1	0	1	1/2	0	3.7081
P09	0	1	3	2/3	8	3.3133
P12	2	5	9	4/7	32	2.0524

602 independent-checker assertions, 245 SymPy identities, clean-process byte replay, and 49/49 repaired-hash hostile mutation rejections. These counts certify implementation and boundary conventions, not the continuum theorem by enumeration.

5 Conclusion and limitations

The result is one complete coherent-family advance: cubic-root exhaustion, cnoidal and soliton formulas, exact period observables, all declared degenerations, and the physical circle clock. It is not a nonlinear or spectral stability theorem and does not classify arbitrary KdV solutions. Its formulas are classical and re-derived source-locally; no arithmetic or literature-priority conclusion is inferred.

References

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[2] M. Leitner and A. Mikikits-Leitner, "Nonlinear differential identities for cnoidal waves," *Mathematische Nachrichten* 287 (2014), 2040–2056, doi:10.1002/mana.201300233.

[3] P. D. Lax, "Integrals of nonlinear equations of evolution and solitary waves," *Communications on Pure and Applied Mathematics* 21 (1968), 467–490, doi:10.1002/cpa.3160210503.